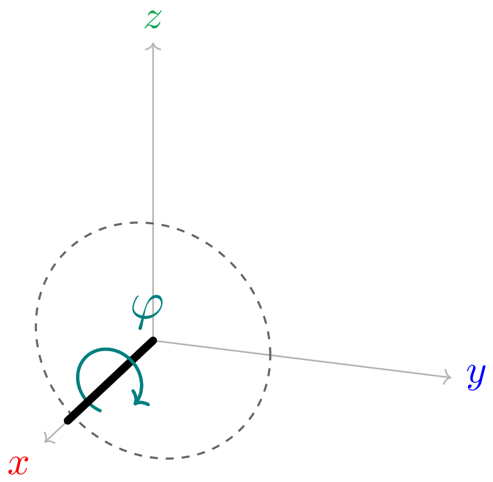
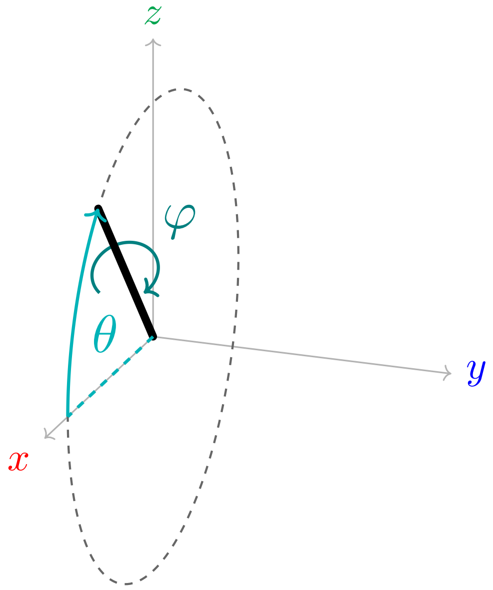
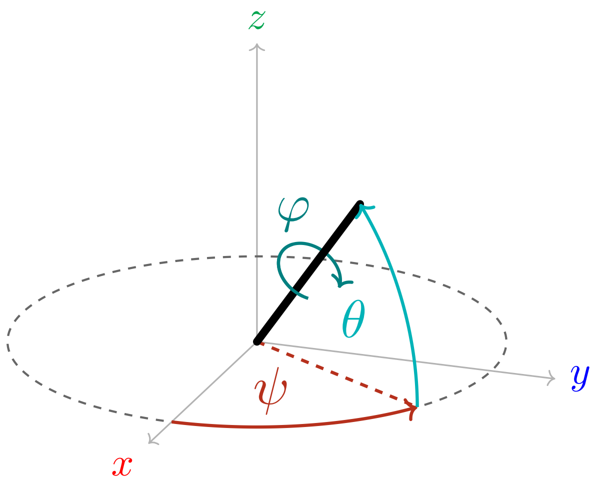
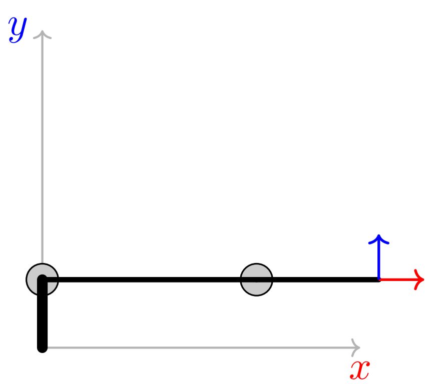
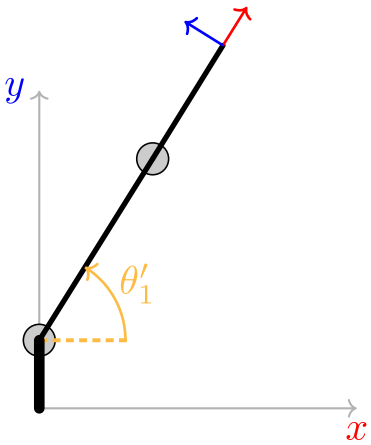
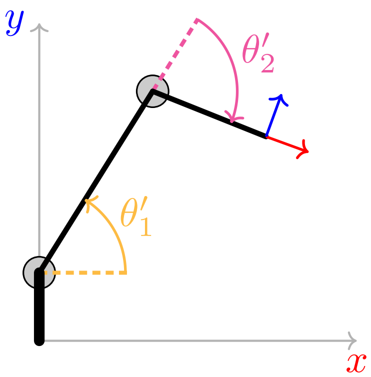

z
φ
y
x

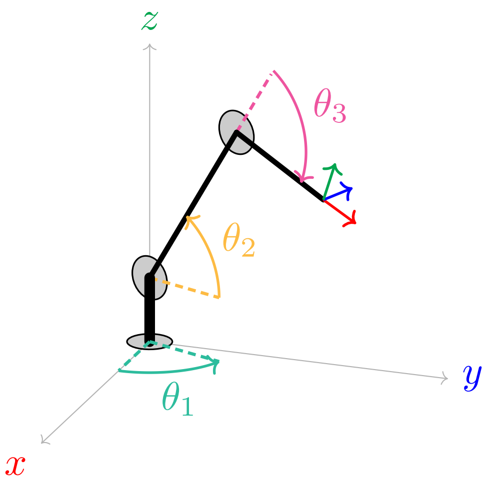

z
φ
θ
y
x

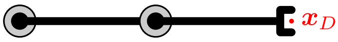

z
φ
θ
y
ψ
x

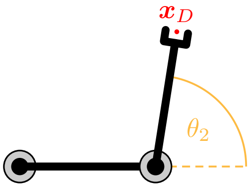

y
x

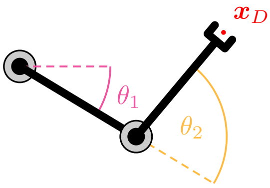

y
θ'_1
x

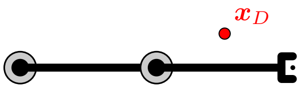

y
θ'_2
θ'_1
x

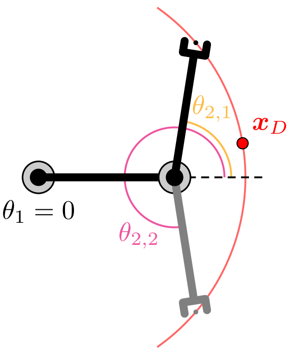

z
θ_3
θ_2
θ_1
y
x

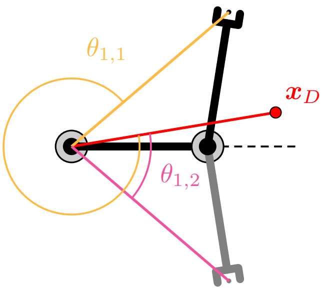












$$L_n = \begin{bmatrix} L_{n,x} \\ L_{n,y} \\ L_{n,z} \end{bmatrix}$$

$$R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$$X_D = L_1 + R(\theta_1)\, L_2 + R(\theta_1 + \theta_2)\, L_3$$
$$X_D = R(\theta_1)\,(L_1 + R(\theta_2)\, L_2)$$

$$x, y, z, \psi, \theta, \phi \longrightarrow \theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6$$
$$\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6 \longrightarrow x, y, z, \psi, \theta, \phi$$